Aspirin

Mechanism of action

- Aspirin (acetylsalicylic acid) has antiplatelet, antipyretic, anti-inflammatory and analgesic effects. In the out-of-hospital setting aspirin is only administered for its antiplatelet activity.
- Aspirin inhibits the enzyme cyclooxygenase which results in a reduction in the formation of prostaglandins and thromboxane.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to 'cardiac chest pain' in the FR Field Guide.

Indications

- ✓ Myocardial ischaemia.
- ✓ STEMI.

Contraindications

- ✕ Known severe allergy.
- ✕ Third trimester of pregnancy.

Cautions

- ⚠ Known bleeding disorder. Aspirin will increase the risk of bleeding, however the balance of risk is usually in favour of administering aspirin.
- ⚠ Clinically significant bleeding. Aspirin will increase the risk of bleeding and the nature and risk of the bleeding must be taken into account.
- ⚠ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs (including aspirin) and a decision must be made based on the balance of risk. If there is a clear history of significant bronchospasm with NSAIDs, aspirin should be withheld.

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Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Aspirin has been associated with premature delivery and premature closure of the ductus arteriosus, when administered in the third trimester of pregnancy.
- The likelihood of clinically important myocardial ischaemia occurring in a woman who is pregnant is so low that the balance of risk is usually in favour of aspirin being withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 300 mg.

Administration

- Administer PO. Dispersible tablets may be chewed or dissolved in water.

Common adverse effects

- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with long-term administration.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 3-5 days for the antiplatelet activity. This is because platelets exposed to aspirin are impaired for the life of the platelet which is 7-10 days. Approximately 10% of platelets are replaced each day.

Usual preparation

- 300 mg dispersible tablets.

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Pharmacokinetics

- Absorption occurs in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Aspirin is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Aspirin displaces warfarin from binding sites and increases the activity of warfarin. However, this effect is most prominent with chronic administration and aspirin is indicated if a patient taking warfarin has clinically significant myocardial ischaemia.

Additional information

- 111 Call Handlers will give instructions for patients to take aspirin if they are suspected of having myocardial ischaemia. This means that some patients that do not have myocardial ischaemia will be advised to take aspirin. The risk for these patients is extremely low and there is potential benefit for those patients who are experiencing myocardial ischaemia.
- The contraindications to aspirin in the call handling protocols are different to those within these CPGs and these differences are not clinically significant.
- Many patients have enteric coated aspirin and not dispersible aspirin and they will be given instructions to chew and swallow their enteric coated aspirin.
- Enteric coated aspirin is not destroyed when it is chewed and will be absorbed. It is however, quite unpleasant to chew and swallow.
  - If the patient still has the tablets in their mouth, ask them to spit the tablets out and administer an additional 300 mg of dispersible aspirin.
  - If the patient has chewed and swallowed 300 mg of aspirin (including enteric coated aspirin), do not administer additional aspirin.
  - If the patient has swallowed (without chewing) 300 mg of enteric coated aspirin, administer an additional 300 mg of dispersible aspirin. This is because absorption of the enteric coated aspirin will be delayed.
  - If it is unclear what the patient has taken, administer an additional 300 mg of dispersible aspirin.
- Urea also impairs platelet function, but aspirin is not contraindicated in the setting of renal failure.

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More Information

Version history

11 September 2023 1.0.4

Update: New indication to align with administrtaion for STEMI.